11.10.2026

Quartile deviation

For groups data,

First quartile range, $Q_1 = \frac{n+1}{4}$,

2nd " " , $Q_3 = \frac{3(n+1)}{4}$

Inter Quartile Range, $IQR = Q_3 - Q_1$

∴ " deviation, $QD = \frac{IQR}{2}$

For group data,

$$Q_1 = L + \frac{\frac{n}{4} - pcf}{f} \times c$$

$$Q_3 = L + \frac{\frac{3n}{4} - pcf}{f} \times c$$

- pcf = previous cumulative frequency

$$IQR = Q_3 - Q_1$$

$$\therefore Q.D = \frac{Q_3 - Q_1}{2}$$

Mean Deviation

For congroup data,

$$M.D_{(mean)} = \frac{\sum (x_i - \bar{x})}{n}$$

- x_i = observations
- $\bar{x}$ = mean
- n = total number of observations.

$$M.D_{(median)} = \frac{\sum_{i=1}^{n} (x_i - median)}{n}$$

$$M.D_{(mode)} = \frac{\sum (x_i - mode)}{n}$$

For ~~group~~ group data:

$$M.D_{(mean)} = \frac{\sum_{i=1}^{n} f_i (x_i - \bar{x})}{n} \quad ; f_i = \text{frequency}$$

$$M.D_{(median)} = \frac{\sum_{i=1}^{n} f_i (x_i - median)}{n}$$

$$M.D_{(mode)} = \frac{\sum f_i (x_i - mode)}{n}$$

Standard Deviation

- Actual method / Direct method
- Assume mean method
- Step deviation method

Actual:

$$\sigma = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n}}$$

} Ungroupe

$x = x_i - \bar{x}$

~~$x = x_i - \bar{x}$~~

$$\sigma = \sqrt{\frac{\Sigma f x^2}{n} - \left(\frac{\Sigma f x}{n}\right)^2}$$

↓

Group data

Assume:

$d = x_i - A$

$$\sigma = \sqrt{\frac{\Sigma f d^2}{n} - \left(\frac{\Sigma f d}{n}\right)^2}$$

Step:

$$d = \frac{x_i - A}{c}$$

$$\sigma = c\sqrt{\frac{\Sigma f d^2}{n} - \left(\frac{\Sigma f d}{n}\right)^2}$$

Mark (x_i)	Student (f_i)	$f_i x_i$	c.f	$x_i - \bar{x}$	x^2	fx^2	fx
10	8	80	8	-20.8	100	800	
20	12	240	20	-10.8	400	4800	
30	20	600	40	-0.8	900	18000	
40	10	400	50	9.2	1600	16000	
50	7	350	57	19.2	2500	17500	
60	3	180	60	29.2	3600	10800	
		$\Sigma f_i x_i =$				$\Sigma fx^2 =$	
	$n = \Sigma f_i = 60$	1850				67900	

$$= \sqrt{\frac{\Sigma fx^2}{n} - \left(\frac{\Sigma fx}{n}\right)^2} = \sqrt{\frac{67900}{60} - \left(\frac{1850}{60}\right)^2}$$

Standard Deviation (S.D) → σ

Actual method / Direct method

$$\sigma = \sqrt{\frac{\Sigma (x_i - \bar{x})^2}{n}}$$

$$\sigma = \sqrt{\frac{\Sigma x^2}{n}}$$

$$\sigma = \sqrt{\frac{\Sigma fx^2}{n} - \left(\frac{\Sigma fx}{n}\right)^2}$$

↓ group data

Assume mean method

$$d = x_i - A$$

$$\sigma = \sqrt{\frac{\Sigma fd^2}{n} - \left(\frac{\Sigma fd}{n}\right)^2}$$

A = Assume mean

Step deviation method

$$d = \frac{x_i - A}{c}$$

$$\sigma = c\sqrt{\frac{\Sigma fd^2}{n} - \left(\frac{\Sigma fd}{n}\right)^2}$$

CamScanner

Hand Shaking Theory: (4 marks)

"The sum of the degrees over all the vertices equals twice the number of edges."

Let $G1 = (V, E)$ be a undirected graph G1 with e edges.

Then,

$$\sum_{v=V} \deg(v) = 2e.$$

(*) How many edges are there in a graph with 20 vertices, each of degree 6?

⇒ There are 10 vertices.

Each of degree = 6.

$\therefore 2e = 10 \times 6 = 60$

$\Rightarrow e =$ We know that,

$$\sum_{v \in V} \deg(v) = 2e$$

$\Rightarrow 6 \times 10 = 2e$

$\Rightarrow 2e = \frac{60}{2}$

$\Rightarrow e = \frac{60}{2}$

$= 30$ (Ans)